DESCRIBING THE GEOMETRY OF INTRUSIVE IGNEOUS BODIES

What is the shape of the intrusion?

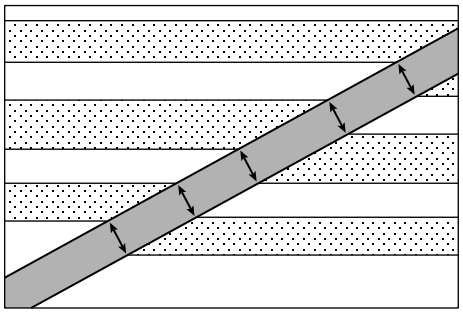
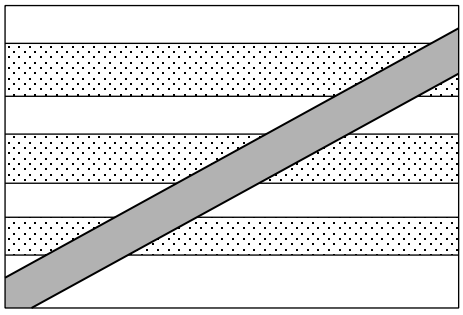
• Tabular: concordant (***sill***) or discordant (***dyke***)
• Arcuate and parallel-sided: inwardly dipping (***cone-sheet***) or vertical / outwardly dipping (***ring dyke***)
• Of more complex shape:
  − ***batholith***: discordant, composite, outcrop area > 100 km^2, no observable bottom
  − ***stock***: discordant, outcrop area < 100 km^2 OR discordant, circular outcrop < 20 km^2
  − ***boss***: discordant, circular outcrop < 25 km^2
  − ***cupola***: discordant, dome-like protuberance from a larger intrusion but isolated from it in map view
  − ***plug***: discordant, vertical cylindrical feed channel of volcano
  − ***diatreme***: discordant, deep, carrot-shaped volcanic vent infilled with breccia
  − ***laccolith***: concordant, dome-like with flat floor
  − ***lopolith***: concordant, saucer-shaped, concave-upward
  − ***phacolith***: concordant, concavo-convex

What are the orientations of the contacts?

• Be quantitative

What is the size of the intrusion?

• Be quantitative

Can anything be said about the mechanism of emplacement?

• An apparent offset of layers in the country rock across the body implies that the intrusion was emplaced by dilation of the country rock; for tabular intrusions the opening direction can be established from two apparent offsets
• No apparent offset generally implies (except in special cases) intrusion by replacement or stoping